ROLL NO: | | | | | | | | | | |

**(To fill the roll no is compulsory)**

INVERTIS UNIVERSITY BAREILLY

# EVEN SEMESTER EXAMINATION - 2024-25

## B.Tech. (CSE)(AI)

## Semester: VIII

## SUBJECT CODE: BCSAI804

## BACKEND DEVELOPMENT

**Time: 03:00 Hrs.** **Maximum Marks: 50**

**NOTE: THE QUESTION PAPER CONTAINS THREE SECTIONS. ATTEMPT ALL THE SECTIONS.**

## SECTION – A

**NOTE: Attempt all parts of the following question. (10 Marks)**

**Q. 1. Answer the following questions: (1 × 5 = 5 Marks)**

**(i)** How to install package in node.js?

**(ii)** How to create the collections in MongoDB?

**(iii)** Write the method to insert the multiple data in MongoDB.

**(iv)** What is the purpose of express.json() in node.js?

**(v)** Which method is use to create the schema in mongoose?

**Q. 2. Answer the following questions: (1 × 5 = 5 Marks)**

**(i)** What is the purpose of the localField and foreignField operator in MongoDB aggregation?

**(ii)** What is difference between findOne () and find ()?

**(iii)** Which method of mongoose is use to connect MongoDB database?

**(iv)** Which package of node.js is use to convert in hash password?

**(v)** Why use app.use () in node.js?



## SECTION – B

**NOTE:Attempt all questions. (5×2×2 = 20 Marks)**

**Q.3.(i) (a)** How to create API route in node.js? Write the code.

**OR**

**(b)** Explain the API HTTP methods.

**(ii) (a)** What is the difference between express and mongoose in Node.js?

**OR**

**(b)** Which node.js package is used for jwt authentication? Also write the code.

**Q.4.(i) (a)** What is the main purpose of express.js.? How it works?

**OR**

**(b)** How to connect database MongoDB to the server NodeJS? Write the code.

**(ii) (a)** Explain the file handling in node.js.

**OR**

**(b)** Discuss asynchronous programming in JavaScript. Explain concepts like callbacks, Promises, and async/await, and describe how they are used to handle asynchronous code.

## SECTION – C

**NOTE: Attempt all questions. (10 × 2 = 20 Marks)**

**Q.5. (a)** Explain the error handling in node.js.

**OR**

**(b)** What is the jwt authentication? Explain in detail.

**Q.6. (a)** What is the main purpose of mongoose.js? Explain its use.

**OR**

**(b)** What is the purpose of Request and Response parameter? Explain in detail.

**(02)**